Descriptive statistics CHEAT SHEET

DATA

element: what the data describe (an order)

variable: a characteristic (region, order value)

observation: one element's values (a row)

5 orders × 5 variables = 25 values

categorical: groups · quantitative: numbers

discrete (counts) · continuous (measurements)

cross-sectional: one time · time series: over time

observational study: record · experiment: assign at random

SCALES OF MEASUREMENT

nominal: labels (region) = ≠

ordinal: ordered labels (tier) + < >

interval: unit, arbitrary 0 (year) + differences

ratio: unit, true 0 ($, days) + ratios

nominal, ordinal → categorical; interval, ratio → quantitative

POPULATION AND SAMPLE

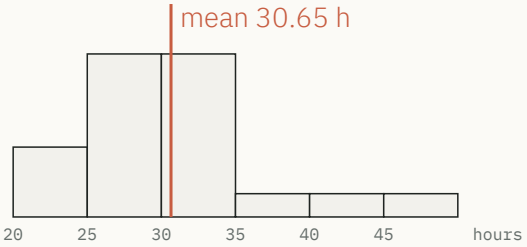
population: all elements · sample: a subset · census: all

parameter (μ, σ) describes a population

statistic ($\bar{x}$, s) describes a sample; an estimator of the parameter

300 of 500 sampled = 60 % → about 60 % of all (inference)

ONE VARIABLE: TABLES AND CHARTS



relative frequency = $\frac{\text{frequency}}{n}$ percent = relative × 100

returns 14, 9, 6, 6, 5 of 40 → 35, 22.5, 15, 15, 12.5 %

pie: 1 % = 3.6°; 35 % → 126° (few parts, never 3-D)

bars: gaps, axis from 0, sorted for nominal data

class width ≈ $\frac{\max - \min}{\text{classes}}$, rounded up: $\frac{47 - 22}{5} = 5$

shipments (h): 20–24 · 25–29 · … · 45–49 → 3, 7, 7, 1, 1, 1

histogram: bars touch; shape: symmetric · skewed right · skewed left

cumulative %: 15, 50, 85, 90, 95, 100 (85 % within 34 h)

stem | leaf: 2 | 2 3 4 5 6 6 7 8 9 9 · 3 | 0 1 2 3 4 4 4 8 · 4 | 1 7

TWO VARIABLES

crosstab: row % compares groups of different size

phone 30/600 = 5 % · laptop 27/300 = 9 % · tablet 5/100 = 5 %

Simpson's paradox: Arrow wins short (90 > 85 %) and long (70 > 60 %), loses overall (76.7 < 80 %): its mix is 2/3 long. Compare within groups.

scatter: positive · negative · none; trendline = direction

side-by-side bars: counts · stacked 100 % bars: mix

honest charts: bars from 0 (axis at 75 % made 3.3 vs 5 look × 3)

scale areas, not heights (2 × tall and wide = 4 × area)

title says what it shows · axes and units · no 3-D · line for time

Descriptive statistics CHEAT SHEET

LOCATION

$\bar{x} = \frac{\Sigma x_i}{n}$ $\mu = \frac{\Sigma x_i}{N}$ Northline: $\frac{120}{12} = 10$ days

weighted: $\bar{x} = \frac{\Sigma w_i x_i}{\Sigma w_i}$ portfolio $0.6 \cdot 8 + 0.3 \cdot 4 + 0.1 \cdot 2 = 6.2$ %

median: the middle value (n even: mean of the middle two)

salaries 38 … 300 ($k): mean 82, median 47 (use the median)

mode: most frequent (Eastgate 7 days; reason 'wrong size')

geometric: $\bar{x}_g = (x_1 x_2 \ldots x_n)^{1/n}$ $\times 2, \times \frac{1}{2} \rightarrow 0$ %, not +25 %

$1.21 \cdot 1 \cdot 1.1 \cdot 1.1 = 1.4641 \rightarrow 1.1$: 10 % a year

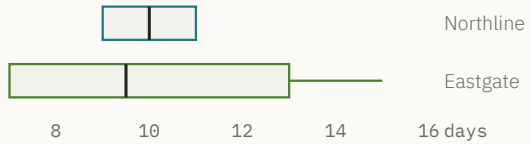
percentile position $L_p = \frac{p}{100}(n + 1)$, interpolate

shipments: $P_{80} = 34$, $P_{90} = 38 + 0.9 \times 3 = 40.7$ h

$Q_1 = P_{25} = 26 \cdot Q_2 = \text{median} = 29.5 \cdot Q_3 = P_{75} = 34$

spreadsheets: .EXC = this rule; .INC differs ($P_{90} = 38.3$)

DISPERSION



range = max − min: 2 and 8 days IQR = $Q_3 - Q_1$: 2 and 6

$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n - 1}$ 2, 4, 4, 5, 10: $\frac{36}{4} = 9$, $s = 3$

$\sigma^2 = \frac{\Sigma(x_i - \mu)^2}{N}$ $n - 1$: deviations sum to 0 ($n - 1$ free)

{1, 3, 5}: $\sigma^2 = 8/3$; 9 samples of 2 → ÷($n - 1$) averages 8/3, ÷n 4/3

suppliers: $s \approx 0.85$ and 2.92 days, both $\bar{x} = 10$

CV = $\frac{s}{\bar{x}} \times 100$ %: 4/6 ≈ 67 %, 6/12 = 50 %

SHAPE, Z-SCORES, OUTLIERS

skew right: mean > median (30.65 > 29.5) · SKEW ≈ 0.98

$z = \frac{x - \bar{x}}{s}$ $(\frac{x - \mu}{\sigma})$ $\frac{52\,000 - 40\,000}{6000} = 2$

Chebyshev: at least $1 - \frac{1}{z^2}$ within z s.d. (z > 1): 75 %, 88.9 %

empirical (bell shape): 68 % · 95 % · 99.7 % within 1, 2, 3 s.d.

rods 250.0 ± 0.4 mm → 249.6–250.4 mm holds about 68 %

outliers: |z| > 3, or below $Q_1 - 1.5$ IQR / above $Q_3 + 1.5$ IQR

shipments: 26 − 12 = 14, 34 + 12 = 46 → 47 flagged (z ≈ 2.6)

five numbers: 22 · 26 · 29.5 · 34 · 47

COVARIANCE AND CORRELATION

$s_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$ sign = direction

km 1–5, minutes 10, 14, 15, 20, 21: $\frac{28}{4} = 7$

$r_{xy} = s_{xy} \div (s_x\, s_y)$ $-1 \le r \le 1$, no units

$\frac{7}{1.58 \times 4.53} \approx 0.98$ returns: $\frac{2.0}{2.08} \approx 0.96$

linear only (an arch can give r = 0) · not cause (confounders)

one point moves r: a 7th day (−0.1 %, +6 %) → 0.96 to 0.52

plot it: the same means, s.d. and r can hide four shapes

SPREADSHEET FUNCTIONS

COUNTIF · FREQUENCY · AVERAGE · SUMPRODUCT/SUM · MEDIAN

MODE.SNGL · GEOMEAN · PERCENTILE.EXC · QUARTILE.EXC

MIN · MAX · VAR.S · STDEV.S · VAR.P · STDEV.P

SKEW · STANDARDIZE · COVARIANCE.S · CORREL